

Abstract

In this talk, I'll start by giving an introduction to the concept of multiplicative sequences in order to define the concept of K -genus of a manifold and use these notions to prove the famous Hirzebruch signature theorem. We will follow very closely what's done in [1], we don't claim any originality for this material. I thank the audience for their very interesting questions.

1 Introduction and notation

Let Λ be a fixed commutative ring with unit (usually it will be $\mathbb{Q}$). The symbol

$$A^* = (A^0, A^1, A^2, \dots)$$

will stand for a graded Λ -algebra with unit which is commutative in the classical sense ($yx = xy$ regardless of the degrees of x and y). In the main application, A^n will be the cohomology group $H^{4n}(B; \Lambda)$.

To each such A^* we associate the commutative ring A^Π consisting of all formal sums $a_0 + a_1 + a_2 + \dots$ with $a_i \in A^i$. We will be particularly interested in the group consisting of all elements of the form

$$a = 1 + a_1 + a_2 + \dots$$

in A^Π . The product of two such units is evidently given by the formula

$$(1 + a_1 + a_2 + \dots)(1 + b_1 + b_2 + \dots) = 1 + (a_1 + b_1) + (a_2 + a_1b_1 + b_2) + \dots$$

Now consider a sequence of polynomials

$$K_1(x_1), K_2(x_1, x_2), K_3(x_1, x_2, x_3), \dots$$

with coefficients in Λ such that, if the variable x_i is assigned degree i , then each $K_n(x_1, \dots, x_n)$ is homogeneous of degree n .

Given A^Π as above, and an element $a \in A^\Pi$ with leading term 1, define a new element $K(a) \in A^\Pi$, also with leading term 1, by the formula

$$K(a) = 1 + K_1(a_1) + K_2(a_1, a_2) + \dots \quad (1)$$

Definition 1.1. The K_n form a *multiplicative sequence* of polynomials if the identity

$$K(ab) = K(a)K(b) \quad (2)$$

is satisfied for all such Λ -algebras A^* and for all $a, b \in A^\Pi$ with leading term 1.

It's important to stress the fact that we require the multiplicativity condition *for all such Λ -algebras A^* and for all $a, b \in A^\Pi$ with leading term 1*. The only thing we fixed in this definition is the coefficient ring Λ . This shouldn't come as a surprise, since we are imposing a property on the polynomials K_n , and not on the product structure of A^Π .

Example 1.2. Given any constant $\lambda \in \Lambda$ the polynomials

$$K_n(x_1, \dots, x_n) = \lambda^n x_n$$

for a multiplicative sequence, with

$$K(1 + a_1 + a_2 + \dots) = 1 + \lambda a_1 + \lambda^2 a_2 + \dots$$

Example 1.3. The identity $K(a) = a^{-1}$ defines a multiplicative sequence with

$$\begin{aligned}
K_1(x_1) &= -x_1 \\
K_2(x_1, x_2) &= x_1^2 - x_2 \\
K_3(x_1, x_2, x_3) &= -x_1^3 + 2x_1x_2 - x_3 \\
K_4(x_1, x_2, x_3, x_4) &= x_1^4 - 3x_1^2x_2 + 2x_1x_3 + x_2^2 - x_4
\end{aligned}$$

And in general

$$K_n = \sum_{i_1+2i_2+\dots+ni_n=n} \frac{(i_1+\dots+i_n)!}{i_1!\dots i_n!} (-x_1)^{i_1} \dots (-x_n)^{i_n}$$

These polynomials can be used to describe the relations between the Pontrjagin classes (or the Chern classes, or the Stiefel-Whitney classes) of two vector bundles with trivial Whitney sum: In fact we have

$$1 = aK(a)$$

which reflects the relation between the classes of two bundles with trivial Whitney sum.

Example 1.4. The polynomials

$$\begin{aligned}
K_{2n+1} &= 0 \\
K_{2n}(x_1, \dots, x_{2n}) &= x_n^2 - 2x_{n-1}x_{n+1} + \dots \mp 2x_1x_{2n-1} \pm 2x_{2n}
\end{aligned}$$

form a multiplicative sequence which can be used to describe the relationship between the (rational) Chern classes of a complex vector bundle ω and the rational Pontrjagin classes of the underlying real bundle $\omega_{\mathbb{R}}$

$$K(c(\omega_{\mathbb{C}})) = p(\omega_{\mathbb{R}})$$

The assumption that we are working with a complex vector bundle can't be dropped, since in general Pontrjagin classes are NOT multiplicative. Instead *we just showed* that for complex vector bundles A, B , we have

$$p(A \oplus B) = K(c(A \oplus B)) = K(c(A)c(B)) = K(c(A))K(c(B)) = p(A)p(B)$$

2 the K -genus and Hirzebruch' signature theorem

The following theorem gives a simple classification of all possible multiplicative sequences. Let A^* be the graded polynomial ring $\Lambda[t]$ where t is an indeterminate of degree 1. Then an element of A^* with leading term 1 can be thought of as a formal power series

$$f(t) = 1 + \lambda_1 t + \lambda_2 t^2 + \lambda_3 t^3 + \dots$$

with coefficients on Λ . In particular $1 + t$ is such an element.

Lemma 2.1 (Hirzebruch). *Given a formal power series $f(t) = 1 + \lambda_1 t + \lambda_2 t^2 + \dots$ with coefficients in Λ , there is one and only one multiplicative sequence $\{K_n\}_n$ with coefficients in Λ satisfying the condition*

$$K(1 + t) = f(t)$$

or equivalently satisfying the condition that the coefficient of x_1^n in each polynomial $K_n(x_1, \dots, x_n)$ is equal to λ_n .

Definition 2.2. $\{K_n\}$ is called the multiplicative sequence *belonging to* the power series $f(t)$.

Example 2.3. The three multiplicative sequences mentioned above belong to the power series $(1 + \lambda t)$, $1 - t + t^2 - t^3 + \dots = \frac{1}{1+t}$, and $1 + t^2$ respectively.

Remark 2.4. If the multiplicative sequence $\{K_n\}$ belongs to the power series $f(t)$, then for any A^* and any $a_1 \in A^1$ the identity $K(1 + a_1) = f(a_1)$ is satisfied. Of course this identity would no longer be true if something of degree $\neq 1$ were substituted in place of a_1 .

Proof of uniqueness. Choosing any positive integer n , let A^* be the polynomial ring $\Lambda[t_1, \dots, t_n]$ where the t_i are algebraically independent of degree 1 and let

$$\sigma = (1 + t_1) \cdots (1 + t_n) \in A^\Pi$$

Then the i -th elementary symmetric polynomial σ_i is the homogeneous part σ of degree i . Hence

$$\sigma = 1 + \sigma_1 + \sigma_2 + \cdots$$

Therefore

$$\begin{aligned} K(\sigma) &= K(1 + \sigma_1 + \sigma_2 + \cdots) \\ &= 1 + K_1(\sigma_1) + K_2(\sigma_1, \sigma_2) + \cdots \end{aligned}$$

On the other hand,

$$\begin{aligned} K(\sigma) &= K\left(\prod_i (1 + t_i)\right) \\ &= K(1 + t_1) \cdots K(1 + t_n) = f(t_1) \cdots f(t_n) \end{aligned}$$

Therefore $K_n(\sigma_1, \dots, \sigma_n)$ is the homogeneous part of $\prod_{i=1}^n f(t_i)$ of degree n . Since $\sigma_1, \dots, \sigma_n$ are algebraically independent, this uniquely determines K_n . $\square$

Proof of existence. For any partition I $i_1, \dots, i_n$ of n , we define $\lambda_I = \lambda_1 \lambda_2 \cdots \lambda_r$. Define

$$K_n(\sigma_1, \dots, \sigma_n) := \sum_I \lambda_I s_I(\sigma_1, \dots, \sigma_n)$$

where we sum over all partitions I of n . Here $s_I(\sigma_1, \dots, \sigma_n)$ is the unique polynomial in the elementary symmetric polynomials equal to $\sum_p t_{\sigma(1)}^{t_1} \cdots t_{\sigma(r)}^{t_r}$ where we sum over all permutations p of $\{1, \dots, r\}$. Define

$$s_I(1 + l_1 t + l_2 t^2 + \cdots) := s_I(l_1 t, l_2 t^2, \dots, l_n t^n)$$

Then

$$s_I(ab) = \sum_{HJ=I} s_J(a) s_H(b)$$

for all $a, b \in (\Lambda[[t]])^\times$. Therefore

$$\begin{aligned} K(ab) &= \sum_I \lambda_I s_I(ab) \\ &= \sum_I \lambda_I \sum_{HJ=I} s_H(a) s_J(b) \\ &= \sum_I \sum_{HJ=I} \lambda_H s_H(a) \lambda_J s_J(b) \\ &= K(a)K(b) \end{aligned}$$

for all $a, b \in (\Lambda[[t]])^\times$. Hence K is multiplicative. The coefficient of σ_1^n of $K_n(\sigma_1, \dots, \sigma_n)$ is λ_n . $\square$

Remark 2.5. One can read this lemma in the other direction, in the following sense. Given a multiplicative sequence K (and here multiplicative is crucial), one can find a formal power series $f(t)$ by just evaluating K on $(1 + t)$ in such a way that K then belongs to f . The property on the coefficient of x_1^n is trivially satisfied since when evaluating $K(1 + x_1)$ the coefficients popping out in the formal power series are the ones of x_1 and its powers.

Now consider some multiplicative sequence of polynomials $K := \{K_n(x_1, \dots, x_n)\}$ with rational coefficients. Let M^m be smooth, compact, oriented m -dimensional manifold.

Definition 2.6. The K -genus $K[M^m]$ is 0 if m is not divisible by 4. If $m = 4n$ then it's equal to the rational number

$$K[M^{4n}] := \langle K_n(p_1(TM), \dots, p_n(TM)), \mu_{4n}^M \rangle \quad (3)$$

Thus $K[M^{4n}]$ is a certain rational linear combination of the Pontrjagin numbers of M^m .

Claim 2.7. *The K -genus is additive under disjoint union of manifolds of the same dimension*

Proof. Clearly the only non-trivial case is when dimension is a multiple of 4. Let M^{4n} and N^{4n} be closed, oriented manifolds:

$$K_n([M^{4n}] + [N^{4n}]) = K_n([M^{4n} \amalg N^{4n}]) \quad (4)$$

$$= \langle K_n(p_1(T(M \amalg N)), \dots, p_n(T(M \amalg N))), \mu_{4n}^{M \amalg N} \rangle \quad (5)$$

$$= \langle K_n(p_1(TM), \dots, p_n(TM)), \mu_{4n}^M \rangle + \langle K_n(p_1(TM), \dots, p_n(TM)), \mu_{4n}^N \rangle \quad (6)$$

$$= K_n([M^{4n}]) + K_n([N^{4n}]) \quad (7)$$

where in (6) we used the fact that $i_M^* p_i(T(M \amalg N)) = p_i(T(M))$ and $i_N^* p_i(T(M \amalg N)) = p_i(T(N))$ together with the fact that $\mu_{4n}^{M \amalg N} = i_{M*} \mu_{4n}^M + i_{N*} \mu_{4n}^N$ and that when we evaluate char classes of M on the fundamental class of N we get zero and vice-versa. $\square$

Claim 2.8. *Assume that $M = \partial W$, then $K[M] = 0$*

Proof. Clearly the only non-trivial case is when $M^{4n} = \partial W^{4n+1}$. Recall that $\mu_{4n}^M = \partial \mu_{4n+1}^W$, where with an abuse of notation we denote with ∂ both the boundary operator and the actual boundary. On the other hand, let $i: M \rightarrow W$ the inclusion, then

$$i^* p_k(TW) = p_k(i^* TW) \quad (8)$$

$$= p_k(TM \oplus \mathbb{R}) \quad (9)$$

$$= p_k(TM) \quad (10)$$

therefore when computing a Pontrjagin number we have:

$$\langle p_1(TM)^{i_1} \cup \dots \cup p_n(TM)^{i_n}, \mu_{4n}^M \rangle = \langle p_1(TM)^{i_1} \cup \dots \cup p_n(TM)^{i_n}, \partial \mu_{4n}^W \rangle \quad (11)$$

$$= \langle i^* p_1(TW)^{i_1} \cup \dots \cup i^* p_n(TW)^{i_n}, \partial \mu_{4n}^W \rangle \quad (12)$$

$$= \langle i^* (p_1(TW)^{i_1} \cup \dots \cup p_n(TW)^{i_n}), \partial \mu_{4n}^W \rangle \quad (13)$$

$$= \langle \delta i^* (p_1(TW)^{i_1} \cup \dots \cup p_n(TW)^{i_n}), \mu_{4n}^W \rangle = 0 \quad (14)$$

where we used naturality of cup product and the fact that δ and ∂ are adjoint. The last passage is by the exactness of the exact sequence of the pair (W, M) . $\square$

Lemma 2.9. *For any multiplicative sequence $\{K_n\}$ with rational coefficients, the correspondence $M \mapsto K[M]$ defines a ring homomorphism from the cobordism ring Ω_* to the rational numbers $\mathbb{Q}$*

Equivalently, this correspondence gives rise to an algebra homomorphism from $\Omega_* \otimes \mathbb{Q}$ to $\mathbb{Q}$.

Proof. The preceding claims shows that it's a morphism of abelian groups from Ω to $\mathbb{Q}$. Let's prove that's a ring homomorphism. For a product manifold $M \times N$ with total Pontrjagin class congruent to $p^M \times p^N$ modulo elements of order 2, we have $K(p^M \times p^N) = K(p^M) \times K(p^N)$ (since it's a multiplicative sequence and we are killing torsion elements), hence

$$\langle K(p^M \times p^N), \mu^M \times \mu^N \rangle = (-1)^{\dim M \dim N} \langle K(p^M), \mu^M \rangle \langle K(p^N), \mu^N \rangle$$

Since the sign in this formula (in the non trivial case) is surely $+1$, this proves that

$$K(M \times N) = K(M)K(N)$$

$\square$

Recall the following properties about the signature σ of a closed oriented manifold

Proposition 2.10. • $\sigma(M \amalg N) = \sigma(M) + \sigma(N)$

- $\sigma(M \times N) = \sigma(M)\sigma(N)$
- if M is an oriented boundary, then $\sigma(M) = 0$.

It should be clear now that the signature might arise as a combination of Pontrjagin numbers. In fact we have the following result.

Theorem 2.11 (Hirzebruch's signature theorem). *Let $\{L_k(p_1, \dots, p_k)\}$ be the multiplicative sequence of polynomials belonging to the power series*

$$\frac{\sqrt{t}}{\tanh(\sqrt{t})} = 1 + \frac{1}{3}t - \frac{1}{45}t^2 + \dots + (-1)^{k-1} \frac{2^{2k} B_k}{(2k)!} t^k + \dots$$

Then the signature $\sigma(M^{4k})$ of any smooth compact oriented manifold M^{4k} is equal to the L -genus $L[M^{4k}]$.

Here B_k denotes the k -th Bernoulli number, with

$$B_1 = \frac{1}{6} \quad B_2 = \frac{1}{30} \quad B_3 = \frac{1}{42} \quad \dots$$

The first four L -polynomials are

$$\begin{aligned} L_1 &= \frac{1}{3}p_1 \\ L_2 &= \frac{1}{45}(7p_2 - p_1^2) \\ L_3 &= \frac{1}{945}(62p_3 - 13p_2p_1 + 2p_1^3) \\ L_4 &= \frac{1}{14175}(381p_4 - 71p_3p_1 - 19p_2^2 + 22p_2p_1^2 - 3p_1^4) \end{aligned}$$

Proof. Since the correspondence $M \mapsto \sigma(M)$ and $M \mapsto L[M]$ both gives rise to algebra homomorphisms from $\Omega_* \otimes \mathbb{Q}$ to $\mathbb{Q}$, it suffices to check this theorem on a set of generators for the algebra $\Omega_* \otimes \mathbb{Q}$. It's known that the complex projective spaces $\mathbb{C}P^{2k}$ provide such a set of generators. To compute the signature of $\mathbb{C}P^{2k}$, we need only note that $H^{2k}(\mathbb{C}P^{2k}; \mathbb{Q})$ is generated by a single element a^k (remember that $|a| = 2$) with

$$\langle a^k \smile a^k, \mu_{4k}^{\mathbb{C}P^{2k}} \rangle = 1$$

Hence $\sigma(\mathbb{C}P^{2k}) = +1$. To compute $L_k[\mathbb{C}P^{2k}]$, we recall that the tangential Pontrjagin class p of $\mathbb{C}P^{2k}$ is equal to $(1 + a^2)^{2k+1}$. Since the multiplicative sequence $\{L_k\}$ belongs to the power series

$$f(t) = \frac{\sqrt{t}}{\tanh(\sqrt{t})}$$

it follows that

$$L(1 + a^2) = \frac{\sqrt{a^2}}{\tanh(\sqrt{a^2})}$$

and hence that

$$L(p) = \left(\frac{a}{\tanh(a)} \right)^{2k+1}$$

Thus the L -genus $\langle L(p), \mu \rangle$ is equal to the coefficients of a^{2k} in this power series. By Cauchy's integral formula this coefficient is equal to

$$\frac{1}{2\pi i} \oint \frac{1}{z^{2k+1}} \frac{z^{2k+1}}{\tanh(z)^{2k+1}} dz = \frac{1}{2\pi i} \oint \frac{1}{\tanh(z)^{2k+1}} dz \quad (15)$$

$$= \frac{1}{2\pi i} \oint \frac{1}{(1-v^2)v^{2k+1}} dv \quad (16)$$

$$= \frac{1}{2\pi i} \oint \frac{\sum_{j=1}^{\infty} v^{2j}}{v^{2k+1}} dv \quad (17)$$

$$= \frac{1}{2\pi i} \oint \frac{\sum_{j=1}^k v^{2j}}{v^{2k+1}} dv + \frac{1}{2\pi i} \oint \frac{\sum_{j=k+1}^{\infty} v^{2j}}{v^{2k+1}} dv = 1 + 0 = 1 \quad (18)$$

where we used in the last passage that $\frac{\sum_{j=k+1}^{\infty} v^{2j}}{v^{2k+1}}$ is holomorphic sufficiently near the origin while $\frac{\sum_{j=1}^k v^{2j}}{v^{2k+1}}$ has residue 1.

Hence $L[\mathbb{C}P^{2k}] = \sigma(\mathbb{C}P^{2k}) = 1$, which implies that $L[M] = \sigma(M)$ for all oriented closed manifold M . $\square$

Corollary 2.12. *The L -genus is always an integer*

Corollary 2.13. *The L -genus is a homotopy invariant of M*

Incredibly enough, according to [2], the L -genus and its rational multiplies are the only rational linear combinations of Pontrjagin numbers which are oriented homotopy type invariants.

References

- [1] John W. Milnor, James D. Stasheff, *Characteristic Classes*, **Annals of Mathematics studies**, no. 76
- [2] P.J. Kahn, *Characteristic Numbers and Oriented Homotopy Type*, **Topology** Vol. 3, pp. 81-95.